



Entropy is not continuous in standard probability spaces

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Abstract

We show that the topologies coming from L^1 , total variation and Jensen-Shannon divergence are all equivalent but do not guarantee the continuity of the entropy. However, a topology that guarantees that mixing (i.e. convex combination) and the entropy are continuous must contain the L^1 /total variation/JSD topology.

1 Introduction

Unlike in the finite-dimensional case, multiple topologies are possible in infinite-dimensional linear/convex spaces. This raises the question as to what is the appropriate one for probability spaces, particularly in the context of ensembles with an entropy defined. The L^1 topology for probability densities, the total variation topology for probability measures and the topology generated by open balls of the Jensen-Shannon divergence can be shown to be all equivalent, but they are not enough to guarantee the continuity of the entropy. That is, a sequence of distributions may converge in total variation while its entropy fails to converge. Conversely, we show that a topology that guarantees the continuity of statistical mixing and the entropy guarantees the convergence in L^1 /total variation/JSD.

2 Definitions and basic properties

Let (X, Σ, μ) be a measure space. Let $S(\rho) = -\int_X \rho \log \rho d\mu$ be the entropy of a probability measure (i.e. ρ is the Radon-Nikodym derivative of a probability measure absolutely continuous with respect to μ). Let $P_\mu = \{\rho \in L^1(\mu) \mid \rho \geq 0, \int_X \rho d\mu = 1, S(\rho) \in \mathbb{R}\}$ be the space of μ -densities with finite entropy.¹

We now want to compare three different convergence criteria that give rise to the same topology. Considering the probability densities as functions, the L^1 distance between densities is

$$d_1(\rho, \sigma) = \|\rho - \sigma\|_{L^1(\mu)} = \int_X |\rho - \sigma| d\mu. \quad (1)$$

¹Logarithms are taken in base 2 unless otherwise stated.

If p_ρ and p_σ are the corresponding probability measures, then the total variation between the probability measures is given by

$$\|p_\rho - p_\sigma\|_{\text{TV}} = \frac{1}{2} \|\rho - \sigma\|_{L^1(\mu)}. \quad (2)$$

Thus L^1 convergence of densities is equivalent to convergence in total variation of the corresponding measures. The Jensen–Shannon divergence between the two densities is given by

$$\text{JSD}(\rho, \sigma) = S\left(\frac{\rho + \sigma}{2}\right) - \frac{1}{2}S(\rho) - \frac{1}{2}S(\sigma). \quad (3)$$

We can show that this too generates the same topology.

Lemma 4. *For $u \in [-1, 1]$, define*

$$f(u) = \frac{1}{2} [(1+u) \log(1+u) + (1-u) \log(1-u)], \quad (5)$$

with the convention $0 \log 0 = 0$. Then

$$\frac{u^2}{2 \ln 2} \leq f(u) \leq |u|. \quad (6)$$

Proof. The function f is even, with $f(0) = 0$ and $f(1) = 1$. For $u \in (-1, 1)$,

$$f''(u) = \frac{1}{(1-u^2) \ln 2}. \quad (7)$$

Therefore $f''(u) \geq \frac{1}{\ln 2}$. Since $f(0) = f'(0) = 0$, this gives

$$f(u) \geq \frac{u^2}{2 \ln 2}. \quad (8)$$

Moreover, $f''(u) \geq 0$, so f is convex. Hence on $[0, 1]$, f lies below the chord joining $(0, 0)$ and $(1, 1)$:

$$f(u) = f((1-u)0 + u1) \leq (1-u)f(0) + uf(1) = u. \quad (9)$$

By evenness, this gives $f(u) \leq |u|$ on all of $[-1, 1]$. $\square$

Proposition 10 (Jensen–Shannon and total variation generate the same topology). *Let $\rho, \sigma \in P_\mu$. Then*

$$\frac{1}{8 \ln 2} \|\rho - \sigma\|_{L^1(\mu)}^2 \leq \text{JSD}(\rho, \sigma) \leq \frac{1}{2} \|\rho - \sigma\|_{L^1(\mu)}. \quad (11)$$

Consequently,

$$\text{JSD}(\rho_n, \rho) \rightarrow 0 \iff \|\rho_n - \rho\|_{L^1(\mu)} \rightarrow 0. \quad (12)$$

Equivalently, the topology generated by the Jensen–Shannon divergence is the total variation topology on the associated probability measures.

Proof. Let

$$m = \frac{\rho + \sigma}{2}. \quad (13)$$

Where $\rho + \sigma > 0$, define

$$u = \frac{\rho - \sigma}{\rho + \sigma}, \quad (14)$$

and set $u = 0$ where $\rho + \sigma = 0$. Then $u \in [-1, 1]$ and

$$\rho = m(1 + u), \quad \sigma = m(1 - u). \quad (15)$$

Substituting into the definition of Jensen–Shannon divergence gives

$$\text{JSD}(\rho, \sigma) = \int_X m f(u) d\mu. \quad (16)$$

By the scalar bounds on f ,

$$\frac{1}{2 \ln 2} \int_X m u^2 d\mu \leq \text{JSD}(\rho, \sigma) \leq \int_X m |u| d\mu. \quad (17)$$

Since $m d\mu$ is a probability measure, Cauchy–Schwarz gives

$$\begin{aligned} \left(\int_X m |u| d\mu \right)^2 &= \left(\int_X \sqrt{m} (\sqrt{m} |u|) d\mu \right)^2 \leq \left(\int_X \sqrt{m}^2 d\mu \right) \left(\int_X (\sqrt{m} |u|)^2 d\mu \right) \\ &= \left(\int_X m d\mu \right) \left(\int_X m u^2 d\mu \right) = \int_X m u^2 d\mu, \end{aligned} \quad (18)$$

Moreover,

$$\int_X m |u| d\mu = \int_X \frac{\rho + \sigma}{2} \frac{|\rho - \sigma|}{\rho + \sigma} d\mu = \frac{1}{2} \|\rho - \sigma\|_{L^1(\mu)}. \quad (19)$$

Putting it all together, we have

$$\begin{aligned} \frac{1}{2 \ln 2} \left(\int_X m |u| d\mu \right)^2 &\leq \text{JSD}(\rho, \sigma) \leq \int_X m |u| d\mu \\ \frac{1}{8 \ln 2} \|\rho - \sigma\|_{L^1(\mu)}^2 &\leq \text{JSD}(\rho, \sigma) \leq \frac{1}{2} \|\rho - \sigma\|_{L^1(\mu)}. \end{aligned} \quad (20)$$

The two inequalities imply the equivalence of Jensen–Shannon convergence and L^1 convergence. Since L^1 convergence of densities is equivalent to total variation convergence of the associated probability measures, the Jensen–Shannon topology is the total variation topology. $\square$

We want to stress that, given the measure-theoretic construction, all these results apply uniformly to both countable discrete spaces with counting measure and continuous dominated measure spaces.

3 Failure of entropy continuity

We now show that, whenever the measure space contains finite-measure sets of arbitrarily large measure, entropy is not continuous with respect to the L^1 topology.

Theorem 21 (Entropy is not L^1 -continuous on spaces with arbitrarily large finite-measure sets).
Let (X, Σ, μ) be a measure space. Suppose there exists a measurable set $B \in \Sigma$ such that

$$0 < \mu(B) < \infty, \quad (22)$$

and a sequence of measurable sets $A_n \in \Sigma$ such that

$$A_n \cap B = \emptyset, \quad 1 < \mu(A_n) < \infty, \quad \mu(A_n) \rightarrow \infty. \quad (23)$$

Then the entropy

$$S(\rho) = - \int_X \rho \log \rho d\mu \quad (24)$$

is not continuous in the $L^1(\mu)$ topology on finite-entropy probability densities.

Proof. For a measurable set U with $0 < \mu(U) < \infty$, define the uniform probability density on U by

$$u_U = \frac{\mathbf{1}_U}{\mu(U)}. \quad (25)$$

Its entropy is

$$S(u_U) = - \int_U \frac{1}{\mu(U)} \log \left(\frac{1}{\mu(U)} \right) d\mu = \log \mu(U). \quad (26)$$

Let

$$\rho = u_B. \quad (27)$$

Then

$$S(\rho) = \log \mu(B). \quad (28)$$

Define

$$\varepsilon_n = \frac{1}{\log \mu(A_n)}. \quad (29)$$

Since $\mu(A_n) \rightarrow \infty$, we have $\varepsilon_n \rightarrow 0$. Passing to a tail of the sequence if necessary, we may assume $\varepsilon_n \in (0, 1)$ for all n .

Now define

$$\rho_n = (1 - \varepsilon_n)u_B + \varepsilon_n u_{A_n}. \quad (30)$$

Since A_n and B are disjoint,

$$\|\rho_n - \rho\|_{L^1} = \varepsilon_n \|u_{A_n}\|_{L^1} + \varepsilon_n \|u_B\|_{L^1} = 2\varepsilon_n \rightarrow 0. \quad (31)$$

Thus $\rho_n \rightarrow \rho$ in $L^1(\mu)$.

On the other hand, since the supports are disjoint, entropy satisfies the exact mixture formula

$$S(\rho_n) = h_2(\varepsilon_n) + (1 - \varepsilon_n)S(u_B) + \varepsilon_n S(u_{A_n}), \quad (32)$$

where

$$h_2(\varepsilon) = -\varepsilon \log \varepsilon - (1 - \varepsilon) \log (1 - \varepsilon). \quad (33)$$

Therefore

$$S(\rho_n) = h_2(\varepsilon_n) + (1 - \varepsilon_n) \log \mu(B) + \varepsilon_n \log \mu(A_n). \quad (34)$$

By construction,

$$\varepsilon_n \log \mu(A_n) = 1. \quad (35)$$

Moreover,

$$h_2(\varepsilon_n) \rightarrow 0 \quad (36)$$

and

$$(1 - \varepsilon_n) \log \mu(B) \rightarrow \log \mu(B). \quad (37)$$

Hence

$$S(\rho_n) \rightarrow \log \mu(B) + 1. \quad (38)$$

But

$$S(\rho) = \log \mu(B). \quad (39)$$

Thus $\rho_n \rightarrow \rho$ in $L^1(\mu)$, while $S(\rho_n) \not\rightarrow S(\rho)$. Hence entropy is not L^1 -continuous. $\square$

Remark. The proof only uses the existence of finite-measure sets of arbitrarily large measure, disjoint from a fixed finite-measure set. In the countable discrete case with counting measure, one may take $B = \{1\}$ and A_n to be finite sets of cardinality tending to infinity, disjoint from B . Then the construction reduces to the standard example where a probability mass ε_n is spread uniformly over many points.

Remark. Choosing instead

$$\varepsilon_n = \frac{1}{\sqrt{\log \mu(A_n)}}, \quad (40)$$

we still have

$$\varepsilon_n \rightarrow 0, \quad (41)$$

and therefore

$$\rho_n \rightarrow \rho \quad \text{in } L^1(\mu). \quad (42)$$

However,

$$\varepsilon_n \log \mu(A_n) = \sqrt{\log \mu(A_n)} \rightarrow \infty. \quad (43)$$

Therefore

$$S(\rho_n) \rightarrow \infty \quad (44)$$

while still

$$\rho_n \rightarrow \rho \quad \text{in } L^1(\mu). \quad (45)$$

Thus arbitrarily small L^1 perturbations can carry arbitrarily large entropy.

4 Entropy and mixing force total variation

We now show that requiring entropy and statistical mixing to be continuous forces the topology to contain the L^1 /total variation/JSD topology.

Theorem 46 (Entropy continuity and mixing continuity force total variation). *Let τ be a topology on P_μ such that:*

- *statistical mixing (e.g. $(\lambda, \rho, \sigma) \mapsto \lambda\rho + (1 - \lambda)\sigma$) is continuous;*
- *entropy (i.e. $S(\rho) = -\int_X \rho \log \rho d\mu$) is continuous.*

Then τ contains the topology generated by the Jensen–Shannon divergence. Consequently, τ contains the total variation topology, equivalently the $L^1(\mu)$ topology on densities.

Proof. For $\rho, \sigma \in P_\mu$, the Jensen–Shannon divergence is

$$\text{JSD}(\rho, \sigma) = S\left(\frac{\rho + \sigma}{2}\right) - \frac{1}{2}S(\rho) - \frac{1}{2}S(\sigma). \quad (47)$$

Since statistical mixing is continuous and entropy is continuous, the map

$$(\rho, \sigma) \mapsto \text{JSD}(\rho, \sigma) \quad (48)$$

is continuous. Therefore τ must contain the topology generated by the Jensen–Shannon divergence which, as shown in Prop. 10, is the same as the total variation topology, equivalently the $L^1(\mu)$ topology on densities. $\square$

5 Conclusion

The L^1 /total variation topology is the natural strong topology for dominated probability measures. It reduces to the ℓ^1 topology in the countable discrete case and to the $L^1(\mu)$ topology for densities in the general dominated case. The Jensen–Shannon divergence generates this same topology. However, entropy is not continuous in this topology in typical infinite-dimensional spaces. Therefore any ensemble topology that requires both continuous statistical mixing and continuous entropy must refine total variation: Jensen–Shannon or L^1 convergence alone is not sufficient.